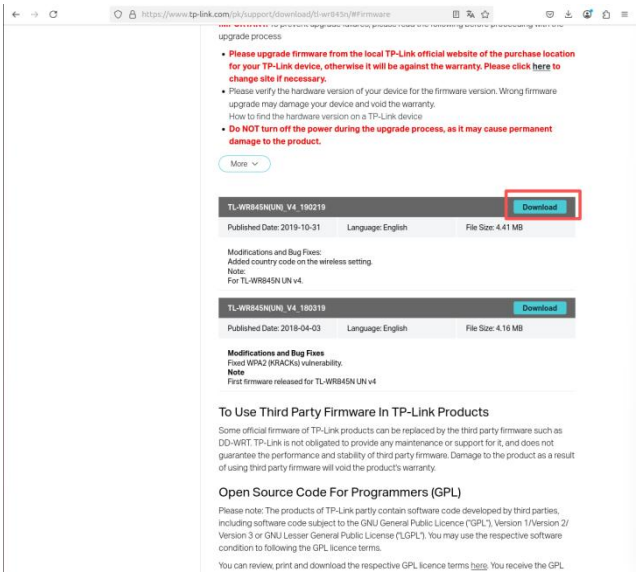
Firmware Analyze

1. Environment

(1) VMware Ubuntu 24.04.2

2. Fireware download

(1) URL:www.tp-link.com/pk/support/download/tl-wr845n/#Firmware
(2) Fireware model: TP-Link TL-WR845N v4



3. Firmware Structure Analysis

(1) Run: binwalk TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin
(2) The output shows:
  ① U-Boot boot information
  ② LZMA compressed segment
  ③ SquashFS file system

```
wyx@wyx-VMware-Virtual-Platform:~/firmware$ binwalk TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
53936         0xD2B0          U-Boot version string, "U-Boot 1.1.3 (Feb 19 2019 - 15:09:38)"
66560         0x10400         LZMA compressed data, properties: 0x5D, dictionary size: 8388608 bytes, uncompressed size: 2953952 bytes
1049088       0x100200        Squashfs filesystem, little endian, version 4.0, compression:xz, size: 2892920 bytes, 544 inodes, blocksize: 262144 bytes,
created: 2019-02-19 07:19:22

wyx@wyx-VMware-Virtual-Platform:~/firmware$
```

4. File system extraction

(1) Run: binwalk -e TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin

(2) Directory successfully extracted:

```
wyx@wyx-VMware-Virtual-Platform:~/firmware$ binwalk -e TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
53936         0xD2B0          U-Boot version string, "U-Boot 1.1.3 (Feb 19 2019 - 15:09:38)"
66560         0x10400         LZMA compressed data, properties: 0x5D, dictionary size: 8388608 bytes, uncompressed size: 2953952 bytes

WARNING: Extractor.execute failed to run external extractor 'sasquatch -p 1 -le -d 'squashfs-root-0' '%e'': [Errno 2] No such file or directory: 'sasquat
ch', 'sasquatch -p 1 -le -d 'squashfs-root-0' '%e'' might not be installed correctly

WARNING: Extractor.execute failed to run external extractor 'sasquatch -p 1 -be -d 'squashfs-root-0' '%e'': [Errno 2] No such file or directory: 'sasquat
ch', 'sasquatch -p 1 -be -d 'squashfs-root-0' '%e'' might not be installed correctly

WARNING: Symlink points outside of the extraction directory: /home/wyx/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashf
s-root/etc/resolv.conf -> /var/tmp/resolv.conf; changing link target to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/wyx/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashf
s-root/etc/TZ -> /var/tmp/TZ; changing link target to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/wyx/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashf
s-root/etc/passwd -> /var/passwd; changing link target to /dev/null for security purposes.
1049088       0x100200        Squashfs filesystem, little endian, version 4.0, compression:xz, size: 2892920 bytes, 544 inodes, blocksize: 262144 bytes,
created: 2019-02-19 07:19:22

wyx@wyx-VMware-Virtual-Platform:~/firmware$ ls
'TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin'  '_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted'
wyx@wyx-VMware-Virtual-Platform:~/firmware$
```

5. File system content analysis

(1) Enter the file system: cd squashfs-root

(2) Visible structure: bin dev etc lib linuxrc mnt proc sbin usr var web

```
wyx@wyx-VMware-Virtual-Platform:~/firmware$ cd _TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted$ ls -al
总计 9636
drwxrwxr-x  4 wyx wyx    4096 11月 21 17:31 .
drwxrwxr-x  3 wyx wyx    4096 11月 21 17:31 ..
-rw-rw-r--  1 wyx wyx 2892920 11月 21 17:31 100200.squashfs
-rw-rw-r--  1 wyx wyx 2953952 11月 21 17:31 10400
-rw-rw-r--  1 wyx wyx 3997184 11月 21 17:31 10400.7z
drwxr-xr-x 13 wyx wyx    4096  2月 19  2019 squashfs-root
drwxrwxr-x  2 wyx wyx    4096 11月 21 17:31 squashfs-root-0
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted$
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted$ cd squashfs-root
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root$ ls -al
总计 52
drwxr-xr-x 13 wyx wyx 4096  2月 19  2019 .
drwxrwxr-x  4 wyx wyx 4096 11月 21 17:31 ..
drwxr-xr-x  2 wyx wyx 4096  2月 19  2019 bin
drwxr-xr-x  5 wyx wyx 4096  2月 19  2019 dev
drwxr-xr-x  5 wyx wyx 4096 11月 21 17:31 etc
drwxr-xr-x  3 wyx wyx 4096  2月 19  2019 lib
lrwxrwxrwx  1 wyx wyx   11  2月 19  2019 linuxrc -> bin/busybox
drwxr-xr-x  2 wyx wyx 4096  7月 16  2018 mnt
drwxr-xr-x  2 wyx wyx 4096  7月 16  2018 proc
drwxr-xr-x  2 wyx wyx 4096  2月 19  2019 sbin
drwxr-xr-x  2 wyx wyx 4096  7月 16  2018 sys
drwxr-xr-x  4 wyx wyx 4096  7月 16  2018 usr
drwxr-xr-x  2 wyx wyx 4096  7月 16  2018 var
drwxr-xr-x  8 wyx wyx 4096  2月 19  2019 web
```

6. Hardcoded password lookup

(1) Enter the etc folder

(2) The passwd file can be found, but it points to null

(3) The passwd.bak file was found

(4) Use cat to view

(5) Obtain administrator account and password

① Administrator account: admin

② Password: $1$5iC.dUsGpxNNJGeOm1dFio/

```
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root$ cd etc
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc$ ls
default_config.xml  init.d          MT7628_AP_2T2R-4L_V15.BIN   passwd.bak              resolv.conf   services          TZ
fstab               inittab         MT7628_EEPROM_20140317.bin  ppp                     RT2860AP.dat  SingleSKU_CE.dat
group               iptables-stop   passwd                      reduced_data_model.xml  samba         SingleSKU_FCC.dat
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc$ ls -al
总计 244
drwxr-xr-x  5 wyx wyx   4096 11月 21 17:31 .
drwxr-xr-x 13 wyx wyx   4096  2月 19  2019 ..
-rwxr-xr-x  1 wyx wyx  52688  2月 19  2019 default_config.xml
-rwxr-xr-x  1 wyx wyx     95  7月 16  2018 fstab
-rwxr-xr-x  1 wyx wyx    877  7月 16  2018 group
drwxr-xr-x  2 wyx wyx   4096  2月 19  2019 init.d
-rw-r--r--  1 wyx wyx     49  7月 16  2018 inittab
-rwxr-xr-x  1 wyx wyx    284  7月 16  2018 iptables-stop
-rw-r--r--  1 wyx wyx    512  2月 19  2019 MT7628_AP_2T2R-4L_V15.BIN
-rwxr-xr-x  1 wyx wyx    512  7月 16  2018 MT7628_EEPROM_20140317.bin
lrwxrwxrwx  1 wyx wyx      9 11月 21 17:31 passwd -> /dev/null
-rw-r--r--  1 wyx wyx    132  7月 16  2018 passwd.bak
drwxr-xr-x  2 wyx wyx   4096  7月 16  2018 ppp
-rw-r--r--  1 wyx wyx 114368  2月 19  2019 reduced_data_model.xml
lrwxrwxrwx  1 wyx wyx      9 11月 21 17:31 resolv.conf -> /dev/null
-rw-r--r--  1 wyx wyx   1949  2月 19  2019 RT2860AP.dat
drwxr-xr-x  2 wyx wyx   4096  7月 16  2018 samba
-rwxr-xr-x  1 wyx wyx  19935  7月 16  2018 services
-rwxr-xr-x  1 wyx wyx   1801  7月 16  2018 SingleSKU_CE.dat
-rwxr-xr-x  1 wyx wyx   1785  7月 16  2018 SingleSKU_FCC.dat
lrwxrwxrwx  1 wyx wyx      9 11月 21 17:31 TZ -> /dev/null
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc$ cat passwd.bak
admin:$1$5iC.dUsGpxNNJGeOm1dFio/:0:0:root:/:/bin/sh
dropbear:x:500:500:dropbear:/var/dropbear:/bin/sh
nobody:*:0:0:nobody:/:/bin/sh
```

7. Startup script analysis

(1) View the startup process(rcS)

(2) Get fragment: cp -p /etc/passwd.bak /var/passwd

① This means that the system automatically restores the hard-coded password every time it starts up, and the password cannot be changed through normal means.

② And it can be confirmed that the previous password was indeed the administrator password.

```
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc$ cd init.d
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc/init.d$ ls
rcS
wyx@wyx-VMware-Virtual-Platform:~/firmware/_TL-WR845Nv4_UN_0.9.1_3.16_up_boot[190219-rel55060].bin.extracted/squashfs-root/etc/init.d$ cat rcS
```

```
cp -p /etc/passwd.bak /var/passwd
```